

Comparison of Zigbee, Z-Wave, Wi-Fi, and Bluetooth Wireless Technologies Used in Home Automation

Salim Jibrin Danbatta
Department of Software Engineering
Firat University
Elazig, Turkey
salimdambatta@gmail.com

Asaf Varol
Department of Software Engineering
Firat University
Elazig, Turkey
varol.asaf@gmail.com

Abstract— Most of the home automation wireless technologies on the internet of things are based on ZigBee, Z-Wave, Wi-Fi, and Bluetooth wireless technologies. While these solutions are good enough, users of smart homes are facing challenges of selecting the best technology. This paper compares the performance of ZigBee, Z-Wave Wi-Fi and Bluetooth technologies from the user's perspective. Power consumption, range, cost, ease of use, scalability, and interoperability are the six indices developed as guidelines to potential users, and hence, the results of this study. With this knowledge, the users can make the appropriate choice of wireless technology solution to use in their home automation systems.

Keywords—internet of things, home automation, wireless technologies, Zigbee, Z-Wave, Wi-Fi, Bluetooth

I. INTRODUCTION

Technological trends in connectivity are moving towards wireless solutions. Home automation involves connecting different home appliances together, and controlling them by using control devices such as Arduino microcontroller, which can communicate via Android applications. In wireless-based home automation system, different types of technologies such as Jakovlje et al. [1] ZigBee, Pawar et al. [2] Wi-Fi and Ganguly et al. [3] Bluetooth are been used, with each technology having its own advantages and disadvantages.

When it comes to home automation systems, ZigBee, Z-Wave, Wi-Fi, and Bluetooth are the current technologies we use for communication as noted by Ryu [4], while Smartphone and Computers (like laptop & tablet) are the devices we use for controlling appliances of the smart home itself. The challenge is getting the technology right. For example, while some users are concerned with a wide range of connectivity some are more interested in a low-cost solution. This study presents the comparison of ZigBee, Z-Wave, Wi-Fi, and Bluetooth wireless technologies with their various characteristics. In addition, recommendations are made based on the constraints of power consumption, range, cost, ease of use, scalability, and interoperability.

A literature review is presented in section II, home automation wireless technologies are discussed in section III. The research design is described in section IV while results and discussion are presented in section V, and section VI describes the conclusions of the study.

II. LITERATURE REVIEW

Several studies about home automation systems have been done in the past. They include experimental, descriptive, and comparative analysis involving different wired and wireless technologies used in Home Automation.

Sikandar et al. [5] show how SMS (short message service) based and wireless technology enhance the controlling of home appliances remotely with the aim of providing home security when the user is away from the place. It was experimental research that investigated the work of [2] and [6] and implemented a pseudo code for HACS (home appliances control system). The pseudo code implemented HACS system and turns out to have many advantages such as remote controlling of home appliances, availability and ease of use to the users. However, though, we opine that there cannot be 100% confidence in such a system, because its functionality is based on GSM (global system for mobile communication) technology which is vulnerable to power failure. Notwithstanding, power disruption can be avoided by attaching a battery voltage to the system as a backup power source.

Another descriptive research [7] discusses various literature relating to smart home automation systems and internet of things (IoT) environment with a focus on home security and user privacy risks. Also risk analysis of smart home automation system in the areas of security, industry, and privacy based approaches were designed and conducted.

Based on the literature reviewed, 32 risks were pointed out, from which 4 were classified as severe and 19 as moderate. Human behavior and software component were listed as the source of the severe risks. The major conclusion to this study was that security and privacy should be integrated at the design

stage. However, it is pertinent to note that, information generated in smart home systems is personal in nature thus, risks to user privacy need further specifications because exposure of this privacy has the potential of becoming a breach to the home itself.

Bhatt and Verma [8] showed the outcome of a postgraduate level project that designed and developed building automation systems based on wired network technology. In the research a network diagram, system operation and flowchart of operation were used to come up with a wired home security system that can react by sounding an alarm, change display of LCD keypad or send voice alert to phone in the event of (i) detection of movement, fire or leakage of gas, (ii) opening and closing of door or window, or breakage of glass (iii) emergency help request by pressing panic button or foot switch. The major outcome of this research was the development of a web-based virtual instrument to run a couple of local or remote monitoring and control of buildings. However, the system is entirely based on wired networks, which does not match with the current trend of technology (wireless) that the world is now moving towards. We will suggest that a future improvement of this work should provide an alternative wireless solution that can go along with the wired system.

Similarly, [9] uses smartphones and wireless network technology to provide remote access and to control home appliances effectively. The experimental research showed how by the means of a simple algorithm a useful behavior of home appliance can be archived and human efforts reduced. Arduino UNO (UNO means one in Italian) was used to design the system, and the user interface was also designed as an Android application. In case of intrusion, the electronic chip is able to send a pop-up message to the Android application using the internet. However, internet service does not normally give us 100% confidence due to service interruption or power failure, thus, we will suggest that the system should add another capability like direct SMS to the user's phone number.

Ryu et al. [4] on the other hand describes how the process control system (PCS) security must be improved. It points out how supervisory, control, and acquisition (SCADA) system depends so much on Internet open protocols. The critical infrastructures discussed in the study were the SCADA process of the natural gas control system of the United States, and the ubiquitous sensor network of Korea.

Considering that SCADA was first designed in the 1960s, and at that time security was of less concern, up till today there has been only a physical security measure in place where SCADA was process a system operated in a closed network. The study recommended that the encryption algorithm and intrusion detection system need to be developed for both the control system, and the programmable logic controller respectively.

When it comes to home automation systems, WiFi, ZigBee, and Bluetooth are current technologies which we use for communication, while Smartphone and Computers (like laptop & tablet) are the devices that we use for controlling appliances of the smart home itself. Waghmare [10] performed a comparative analysis of various smart home systems built on these technologies and controlling devices.

The technologies were categorized into (i) web browser (ii) cloud server and (iii) GSM/Bluetooth module. The study finds out that in a web browser the drawbacks are its need for internet connection and also its not being efficient in rural areas (with an estimated cost of \$20,000). In the cloud server, the problem is the maintenance of cloud and continuous internet connection needed (with an estimated cost of \$25,000). Finally, GSM/Bluetooth module is hard to install and has a less area of operation (with an estimated cost of \$20,000).

Another survey research work by [11], employs the use of activity recognition in smart home and intelligent environment. It modeled and implements five activities of the user, i.e. sitting, standing, walking, running and lying down. The study uses extraction set for 3-d features that include human silhouette and motion characteristics. Video recordings for these five activities were used as input to implement the home automation system that automatically controls real-time analysis of human actions. Nevertheless, showdown projection and illumination were the problems solved using fuzzy logic.

Research work on Bluetooth based sophisticated home automation system using a smartphone in [3] presents a smart home automation system based on Arduino microcontroller and Android application. The system is using Bluetooth as the communication medium. In addition to the usual switching on/off of different devices in home automation setups, the experimental research introduced light intensity control using Smartphone and Bluetooth technologies. Block diagram and flowchart of system operations were used to design the implemented system, which turns out to utilize the Bluetooth module for fast and reliable communications between the remote user and home devices. However, while using Bluetooth technology is cost-effective, internet connection provides a wider area of coverage. In view of this, communication via the internet will increase the coverage area when compared to Bluetooth.

Another work [12] implements a ruled based service oriented monitoring system. The system evaluates smart home automation rules in real time, and stores their results as satisfied, temporarily violated or permanently violated, based on the result received from the controller device. NFC (near field communication) reader for NFC authentication and switch sensor were the two services that were implemented by the system. The NFC reader was placed at the entrance, and the switch sensor records opening of the front door, while the Arduino controller sounds an alarm if the door was opened without authentication. The lesson was that the rule based monitoring system is adequate for this kind of environment.

The wi-fi-based home automation system study was carried in [1] to compared the performance of ZigBee solution and device based on WiFi technology. It was a comparative research study that investigates how existing devices are mostly based on ZigBee technologies in home automation. The major conclusion to this study was those home automation solutions are easier to be implemented with Wi-fi technology than using ZigBee technologies with Wi-Fi having less communication latency when compared to ZigBee. And on the other hand, ZigBee is more suitable in a network with many devices. Though, ZigBee technologies are proven to be

sufficient in implementing a home automation system. They are based on mesh topologies and need additional hardware and routers to provide the user with better connectivity. As such, these systems turn out to be very expensive.

A web application that integrates cameras and motion sensor controlled through Raspberry Pi using internet of things was presented in [13]. The system was designed in such a way that with Internet access using Raspberry Pi, a user can control home appliances with a mobile phone or laptop computer. The simple recursive algorithm of the surveillance system starts by checking the status of the home appliances connected to the sensors, then updates the various status into Raspberry Pi database. If the high use of power is detected then the user is notified for further necessary actions, else the system restarts its monitoring process. The system turns out to reduce human interaction and it is affordable with even a possibility of integrating other home appliances like energy monitoring.

[14] presents a comparative analysis of different SmartPlug energy monitoring and control systems and implements a web-based application that allows its users to log in with their credentials. The system which includes Raspberry Pi and Arduino together with sensors that are responsible for connecting all the devices was successful in monitoring and controlling devices in real time. The web-based application presents devices power consumption results graphically and provides the user with the capability of turning on/off devices remotely.

Abrishambaf et al. [15] employed IEC (international electrical commission) 61499 function block standard for a distributed control system. The work implements an architectural solution in the area of flexibility and reconfigurability control in home automation as a substitute for the traditional system of central control in home automation. In addition to this, wireless sensor network was used as the system infrastructure to implements IoT technology, which provides a solution regarding distributed control and monitoring. However, there was no result discussion in the work. The only report was “Contiki-OS have been used to implement a fully IoT compliant system”. As such, more discussion will depend on future work within the context of the study.

A user-friendly home automation system was presented by [16]. Using Bluetooth technology, a smartphone application provides the user with the ability to control 18 devices, which includes home appliances and sensors. A basic flowchart diagram was used to design the proposed system. The algorithm presented in the flowchart starts by establishing a Bluetooth connection and sending commands from a Smartphone. If the command is valid the user will be asked to control or monitor a device, else check the commands again. The system provides a general approach for home automation and has better performance when compared to the current existing Bluetooth systems, thus, not necessarily suitable for elderly or handicapped people.

Contrary to the most home automation systems which are implemented in the English language, [17] presents home automation solution implemented in the Pashto language from northern Pakistan. The method was simulated using MATLAB

and implemented in Arduino, with Mel-frequency cepstral coefficient (MFCC) used for feature extraction and K-Nearest Neighbor (KNN) as a classifier. However, even though this is a new system, its overall accuracy was 78.8%, and statistically this confidence level is not enough. The use of silence removal algorithm will very much help to increase the accuracy and confidence level in this relatively new system.

III. WIRELESS TECHNOLOGIES USED FOR HOME AUTOMATION ON INTERNET OF THINGS

Choosing a network protocol is the first step in obtaining a home automation application. Popular wireless technologies for home automation systems include Zigbee, Wi-Fi, Bluetooth, and Z-Wave as contained in [18].

ZigBee has three radio bands that operate in 868MHz, 915MHz, and 2.4GHz with low-power consumption and range of 300 meters. It can accommodate up to 6000 devices, but its main limitation is its interoperability with devices from different manufacturers, and has a relatively low cost of installation as explained in [19].

Z-Wave was described by [19] as a new wireless home automation technology that uses very low power. It communicates in the frequency of 900MHz and range of around 30 meters. It makes devices double as repeaters and also has network reliability that enables commercial applications [20]. Z-Wave is mostly used for control and monitoring, and it does not allow any interference from Wi-Fi or any other 2.4GHz wireless technology.

Wi-Fi is another wireless technology that is used to connect home automation appliances based on the IEEE 802.11 standards. It operates in 2.4GHz and 5GHz bands which is available worldwide [21]. In home automation, Wi-Fi is mostly used for wireless monitoring and management of home appliances such as power outlets, door, surveillance, and alarms [21].

Bluetooth is a short-range wireless technology based on the IEEE 802.15.1 standard. It presents a very low-cost home automation solution, which uses transceiver microchips in each device [21]. Having a short range of 10 meters, Bluetooth requires low power, and since they already exist in many devices, Bluetooth is very easily accessible for wireless control.

A common list of devices used on the internet of things (IoT) were noted by [21]. The listed devices are particularly for the purpose of home automation. Some of these devices and their wireless technologies are listed in Table I.

TABLE I. TYPICAL IOT DEVICES WIRELESS TECHNOLOGY FOR HOME AUTOMATION

Device	RAM	ROM/FLASH	Power Supply	Wireless Technology
iPhone	2 Gb	up to 128 Gb	battery	WiFi, Bluetooth
Samsung Smart TV	1 Gb	-	mains-powered	WiFi, Z-Wave
Net Smoke Detector	16 Kb	128 Kb	battery	WiFi
Fitbit Smart Wrist Band	16 Kb	128 Kb	battery	Bluetooth LE
Philips Hue Lightbulb	8 Kb	up to 256 Kb	battery	ZigBee
Sensor Devices	4 – 16 Kb	16 – 128 Kb	battery	ZigBee, WiFi, Bluetooth

IV. RESEARCH DESIGN

Most of the literature presented in section II rely on wired or wireless communication between different Internet of Things (IoT) devices. However, these communication media have their different advantages and limitations as we have discussed in section II.

As noted by [21], there are a number of different wireless technologies used for implementing home automation system on the Internet of Things, the problem is how to compare them. Such a comparison always depends on the type of users and their choice of interest. For example, some users are new, while some users have different devices from different manufacturers. And some users want to spend as little as possible. This study investigates wireless technologies used in home automation, based on the constraints of power consumption, range, cost, ease of use, scalability, and interoperability to determine which of them is more suitable for home automation. Nevertheless, some common typical devices for IoT and particularly for home automation are provided in Table I.

To achieve the comparison of wireless technologies used in home automation (which is the main objective of this study), the following strategies were followed:

- (1) Literature review to understand the different technologies involved in home automation systems.
- (2) Comparison of wireless technologies used in home automation using the indices of power consumption, range, cost, ease of use, scalability, and interoperability.
- (3) Comparative analysis of results based on the research indices.

V. RESULTS AND DISCUSSION

In terms of power consumption, Z-Wave, Bluetooth and ZigBee are having a low power consumption of 1mw, 10mw, and 100mw respectively. While Wi-Fi is having high power consumption. Yet, regarding range Wi-Fi has a range of up to

1000m, with Z-Wave having 30m, ZigBee 100m, and Bluetooth with the lowest having 10m. However, when cost is the constraint of interest Bluetooth is the wisest choice with ZigBee, Wi-Fi, and Z-Wave following respectively.

In the situation where many devices (scalability) from a different manufacturer are involved (where interoperability could be an issue), Z-Wave is the best because it allows commercial devices and hundreds of nodes with high density. While at the extreme end of the spectrum, ZigBee does not accommodate devices from other manufacturers. Wi-Fi and Bluetooth are relatively easy to work with since we can connect most of the devices on the internet of things easily to a Wi-Fi. Bluetooth is also easily accessible, as it exists in many electronic devices like mobile phones and others.

TABLE II. COMPARISON OF WIRELESS TECHNOLOGIES FOR HOME AUTOMATION

Indices	ZigBee	Z-Wave	Wi-Fi	Bluetooth
Power consumption	100 mw	1 mw	High	10 mw
Range	100 m	30 m	1000 m	10 m
Cost	Low	High	Medium	Very low
Scalability	6000	> 6000	32	20
Interoperability	Same manufacturer	Different manufacturers	Wi-Fi compatible devices	Bluetooth compatible devices

VI. CONCLUSION

The user's selection of suitable wireless technology for home automation is dependent on the requirement of the selected wireless solution. Z-Wave is the best choice if the variable of interest is energy saving, or the user is interested in using different devices from different manufacturers, or even commercial devices. When the variable of interest is using devices from the same manufacturer with relatively low-cost and power consumption, ZigBee is the best fit in such a situation. In the event ease of use and range are the interest of the user, then the Wi-Fi solution is the most appropriate. However, if cost is the issue, then Bluetooth would provide a better solution. In conclusion, Z-Wave is the recommended home automation wireless technology solution as a result of the findings of this study because it meets all the requisites for creating an ideal smart home such as ease of installation, low power consumption, interoperability, and affordability.

REFERENCES

- [1] S. Jakovljevic, M. Subotic, and I. Papp, "Realisation of a Smart Plug device based on Wi-Fi technology for use in home automation systems," *2017 IEEE Int. Conf. Consum. Electron. ICCE 2017*, pp. 327–328,

2017.

- [2] P. N. Pawar, S. Ramachandran, N. P. Singh, V. V. Wagh, and B. E. Student, "A Home Automation System using the Internet of Things," *Int. J. Innov. Res. Comput. Commun. Eng. (An ISO)*, vol. 3297, 2007.
- [3] S. Das, S. Ganguly, S. Ghosh, R. Sarker, and D. Sengupta, "A Bluetooth Based Sophisticated Home Automation System Using Smartphone," pp. 236–240, 2016.
- [4] D. H. Ryu, H. Kim, and K. Um, "Journal of Loss Prevention in the Process Industries Reducing security vulnerabilities for critical infrastructure," *J. Loss Prev. Process Ind.*, vol. 22, no. 6, pp. 1020–1024, 2009.
- [5] M. Sikandar, H. Khiyal, A. Khan, and E. Shehzadi, "SMS Based Wireless Home Appliance Control System (HACS) for Automating Appliances and Security Preliminaries Home Appliance Control System (HACS)," vol. 6, 2009.
- [6] B. Ciubotaru-Petrescu, D. Chiciudean, R. Cioarga, and D. Stanescu, "Wireless Solutions for Telemetry in Civil Equipment and Infrastructure Monitoring."
- [7] A. Jacobsson, M. Boldt, and B. Carlsson, "A risk analysis of a smart home automation system," *Futur. Gener. Comput. Syst.*, vol. 56, pp. 719–733, 2016.
- [8] J. Bhatt and H. K. Verma, "Design and Development of Wired Building Automation Systems," *Energy Build.*, vol. 103, pp. 396–413, 2015.
- [9] C. Maternaghan *et al.*, "Home automation system using a wireless network," *IEEE Transactions on Consumer Electronics*, vol. 44, no. 4, pp. 153–171, 2014.
- [10] P. Waghmare, M. Chandgude, P. Chaure, and A. Chaudhari, "Survey on Home Automation Systems," pp. 7–10, 2017.
- [11] V. Gobinath, M. G. L, H. H. S, R. Sathiyavathi, and M. A. V, "HOME AND OFFICE AUTOMATION SYSTEM – SURVEY," pp. 435–440, 2017.
- [12] R. Alcarria, D. M. De Andres, B. Bordel, D. S. De Rivera, A. Sanchez-Picot, and T. Robles, "A service-oriented monitoring system based on rule evaluation for Home Automation," *2017 IEEE Int. Conf. Consum. Electron. ICCE 2017*, pp. 329–330, 2017.
- [13] V. Patchava, H. B. Kandala, and P. R. Babu, "A Smart Home Automation technique with Raspberry Pi using IoT," *2015 Int. Conf. Smart Sensors Syst. IC-SSS 2015*, pp. 3–6, 2017.
- [14] A. Chaudhari, B. Rodrigues, and S. More, "Automated IOT based system for home automation and prediction of electricity usage and comparative analysis of various electricity providers: SmartPlug," in *2016 2nd International Conference on Contemporary Computing and Informatics (IC3I)*, 2016, pp. 390–392.
- [15] R. Abrishambaf, M. Bal, and V. Vyatkin, "Distributed home automation system based on IEC61499 function blocks and wireless sensor networks," *Proc. IEEE Int. Conf. Ind. Technol.*, pp. 1354–1359, 2017.
- [16] M. Asadullah and K. Ullah, "Smart home automation system using Bluetooth technology," *ICIEECT 2017 - Int. Conf. Innov. Electr. Eng. Comput. Technol. 2017, Proc.*, 2017.
- [17] S. Nisar, M. Asadullah, and I. S. Member, "Digits Recognition," pp. 0–3, 2017.
- [18] A. J. D. Rathnayaka, V. M. Potdar, and S. J. Kuruppu, "Evaluation of wireless home automation technologies," in *5th IEEE International Conference on Digital Ecosystems and Technologies (IEEE DEST 2011)*, 2011, pp. 76–81.
- [19] "Types of Wireless Communication Technology Used in Home Automation." [Online]. Available: <https://electronicsforu.com/resources/learn-electronics/wireless-technology-types-home-automation>. [Accessed: 17-Nov-2018].
- [20] "Which Home Automation Technology Is Best for You?" [Online]. Available: <https://www.lifewire.com/best-home-automation-technology-817601>. [Accessed: 17-Nov-2018].
- [21] O. Horyachyy, "Comparison of Wireless Communication Technologies used in a Smart Home: Analysis of wireless sensor node based on Arduino in home automation scenario," 2017.